

Yandi Wu

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Department of Mathematics
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Employment	Lovett Instructor Rice University	2024 - Houston, TX
Education	University of Wisconsin, Madison PhD Mathematics, Advisors: Tullia Dymarz, Caglar Uyanik Minor in Computer Science	2018 - 2024 Madison, WI
	University of California, Berkeley BA Mathematics (High Honors)	2014 - 2018 Berkeley, CA
	Budapest Semesters in Mathematics Semester Abroad	Fall 2017 Budapest, Hungary

Research Interests Low-dimensional topology, geometric group theory, hyperbolic geometry, and geometric rigidity theory, including applications to orbifolds, manifolds, and generalizations.

- Publications & Preprints**
1. Expansive flows on 3-manifolds are pseudo-Anosov. (w/ M. Jay, K. Mann, N. Paulet, A. Rechtman, L. Rosales-Ortiz).
In preparation.
 2. Filling knots and essential systole. (w/ C. Leininger).
In preparation.
 3. Balanced Spanning Trees for Triangle Strip Lattices. (w/ S. Dhakal*, L. Nguyen*).
arXiv Preprint, 2026.
 4. Counting and Entropy for Hyperbolic Surface Amalgams. (w/ H. Parlier).
arXiv Preprint, 2026. Submitted.
 5. Filling Links and Essential Systole. (w/ C. Leininger).
arXiv Preprint, 2025. Submitted.
 6. Subactions for Locally CAT(-1) Spaces. (w/ D. Constantine, E. Shrestha).
Erg. Th. Dyn. Sys. 45 (2025), 3619–3662.
 7. Iso-length-spectral Hyperbolic Surface Amalgams.
Groups Geom. Dyn. (2025), 1-18.
 8. Marked Length Spectrum Rigidity for Surface Amalgams.
Trans. Amer. Math. Soc. 378 (2025), 7075-7106.
 9. A Topologically Rigid Set of Quotients of the Davis Complex.
Geom. Dedicata 217 (2023), 1-20.

*denotes an undergraduate author.

Honors & Awards	Excellence in Research Graduate Student Award “For significant and substantial contributions to research”	2023 Madison, WI
	Elizabeth S. Hirschfelder Award “For exceptional research done by a female graduate student”	2022 Madison, WI
	Outstanding Service Award “For special service to the department”	2022 Madison, WI
	Outstanding TA Award “For excellence in the classroom across multiple semesters”	2021 Madison, WI
Invited Talks (Conferences)	AMS Central Fall Sectional Meeting (Duluth, Minnesota) Geometry, group actions, and dynamics	October 2026
	Spring Topology and Dynamics Seminar (Birmingham, AL) Special Session in Geometric Group Theory	Mar 2026
	Joint Math Meetings (Washington, DC) Geometric Group Theory Special Session	Jan 2026
	Midwest Summer School in Geometry, Topology, and Dynamics (Madison, WI)	Jun 2025
	Spring Topology and Dynamics Conference (Newport News, VA) Session on Geometric Topology	March 2025
	Joint Math Meetings (Seattle, WA) Women in Groups, Geometry, and Dynamics Special Session	Jan 2025
	KIAS-Rice Workshop on Geometric Topology (Seoul, Korea)	Sept 2024
	AMS Central Fall Sectional Meeting (San Antonio, TX) Geometric group theory and low dimensional topology	Sept 2024
	AMS Central Spring Sectional Meeting (Milwaukee, Wisconsin) Special session in developments in hyperbolic-like geometry and dynamics	April 2024
	Joint Math Meetings (San Francisco, CA) Session on Geometric Group Theory	Jan 2024 Jan 2024
	World of GroupCraft III (online)	Sept 2023
	Spring Topology and Dynamics Seminar (online) Special Session in Geometric Group Theory	Mar 2023
Invited Talks (Seminars)	University of Luxembourg Geometry & Topology Seminar	June 2026
	The Δ -Seminar: Geometry & Topology in Bern-Neuchâtel-Fribourg	June 2026
	University of Wisconsin, Madison Group Actions & Dynamics Seminar	Mar 2026
	University of Virginia Geometry Seminar	November 2025

	University of Maryland, College Park Topology Seminar	November 2025
	Boston College Geometry, Topology, and Dynamics Seminar	October 2025
	Fribourg Seminars on Analysis & Geometry on Metric Spaces	May 2025
	Yale University Geometry & Topology Seminar	Dec 2024
	Wake Forest Dynamics Seminar	Nov 2024
	Wesleyan University Topology, Geometry, & Dynamics Seminar	Dec 2023
	University of Minnesota, Twin Cities Geometry Seminar	Nov 2023
	Vanderbilt University Topology & Group Theory Seminar	Nov 2023
	Brandeis University Topology Seminar	Nov 2023
	University of Illinois, Chicago Geometry, Topology, & Dynamics Seminar	Nov 2023
	University of Wisconsin, Madison Dynamics Seminar	Oct 2023
	Rice University Topology Seminar	Aug 2023
	University of Wisconsin, Milwaukee Topology Seminar	April 2023
	The Ohio State University Topology and GGT Seminar	Nov 2022
	BSU-Toledo Joint Geometry and Topology Seminar	Sept 2022
Contributed Talks	Underrepresented Students in Topology and Algebra Research Symposium	Mar 2023
Service & Outreach	Organizer, Log Cabin Workshop (March 15-19, 2027): Organize workshop where participants learn about geodesic currents in small groups, and present a minicourses. Organizer, AWM Research Symposium, Groups, Geometry, and Dynamics Special Session (May 16-18, 2025): Organize and invite speakers to talks focusing on work of early career women researchers in geometry, group theory, and dynamics. Member, Rice Math Department Colloquium Committee (Fall 2024 - Spring 2025): Invite and host external speakers for weekly departmental colloquium. Organizer, UW Madison AMS Student Chapter Seminar (Spring 2021 - Spring 2023): Plan and organize weekly department-wide graduate student seminar. Organizer, Gender Minorities in Mathematics At Wisconsin (Fall 2020 - Spring 2022): Organize professional development and social events for gender minorities in the department, coordinate GMMaW colloquiums. Member, UW Madison Committee on TA Policies and Procedures (Spring 2022): Evaluate TAs in the math department, determine policies for TA responsibilities. Referee Work (2024 - present): <i>Groups, Geom., Dyn.</i>	

Mentoring	Summer Undergraduate Research Fellowship (SURF) Project: <i>The mathematics of congressional redistricting</i> Faculty mentor for two undergraduates at Rice	Summer 2026 Houston, TX
	Madison Experimental Mathematics Lab Project: <i>Hyperbolic Geometry and Crochet</i> Graduate mentor for four undergraduate students	Spring 2023 Madison, WI
	Big Ideas in Dynamics Topic: <i>Length functions on currents and applications to dynamics & counting</i> Graduate Mentor for graduate reading group	Fall 2022 Online
	Madison Experimental Mathematics Lab Project: <i>Random Symmetrics of Hyperbolic Space</i> Graduate mentor for four undergraduate students	2022 - 2023 Madison, WI
	Girls' Night Out Project: Mathematics of Epidemics Graduate Mentor for three high school students	Spring 2020 Madison, WI
	Directed Reading Program Topic: <i>Office Hours with a Geometric Group Theorist</i> Graduate mentor for three undergraduate students	Fall 2018 Madison, WI
Outreach Talks	Rice University Undergraduate Colloquium <i>Taffy pullers and surface homeomorphisms</i>	Mar 2026
	SIAM Student Chapter Seminar <i>How I landed my summer internship</i>	Mar 31, 2023 Madison, WI
	UW Madison Math Circle <i>Cut and Paste Topology</i>	Feb 2019, Sept 2021 Madison, WI
Teaching Experience	Instructor, Rice University Math 355 (Linear Algebra) Math 290 (Introduction to Proof Writing) Math 102 (Calculus II)	Houston, TX Spring 2025, Spring 2026 Spring 2025, Fall 2025 Fall 2024, Spring 2026
	Instructor, UW Madison Math 131 (Problem solving in Algebra, Geometry, and Statistics)	Madison, WI Summer 2021
	Teaching Assistant Coordinator Math 234 (Calculus III) Math 222 (Calculus II) Math 221 (Calculus I)	Madison, WI Fall 2021*, Spring 2021*, Fall 2020, Spring 2020* Fall 2022*, Spring 2023* Fall 2019
	Teaching Assistant Math 222 (Calculus and Analytic Geometry II)	Madison, WI Fall 2018, Spring 2019*
	* Received “superior” rating awarded to top 30% of TAs every semester	
Industry Experience	Reddit, Ads Prediction Team	New York, NY

Data Science Intern

Summer 2023

- **Machine Learning:** Implement model calibration techniques to decrease calibration error of decision tree model in production by up to 30 percent.
- **Data Science:** Build Mode dashboards to visualize how different sectors, such as geographic location, ad industry, and user frequency, affect model calibration.
- Present on state-of-the-art model calibration techniques for modern Deep Neural Networks at company-wide Ads Journal Club.

US Army Corps of Engineers, Geospatial Research Lab
NSF Mathematical Sciences Graduate Internship Program

Alexandria, VA
Summer 2022

- **Computer vision:** Implement automated building damage assessment by GPU-powered neural networks on satellite images of natural disaster sites.
- **Transfer Learning:** Implement domain adaptation techniques that increased model accuracy by up to 6 percent.
- White paper, *Transfer Learning Techniques for Building Damage Assessment*, and NSF MSGI Research Symposium slides available on website.